

A Product Counterexample to Problem 5.14(iii)

Strict admissibility does not characterize the three classes in Theorem 4.3

Candidate solution packet for arXiv:1912.03659

27 August 2026

Status. Candidate full negative answer to Problem 5.14(iii), likely valid and awaiting human review. The result answers only part (iii) of the four-part problem.

1 Source question

The source is Karsten Kruse, *Vector-valued Fourier hyperfunctions and boundary values*, arXiv:1912.03659v2, 5 November 2025 [1]. On source PDF page 46, Problem 5.14(iii) asks:

“Is strict admissibility of a sequentially complete E equivalent to belonging to the classes of spaces from Theorem 4.3?”

Theorem 4.3 gives the following three sufficient classes: (a) complex Fréchet spaces; (b) strong duals F'_b of complex Fréchet spaces F satisfying (DN) ; and (c) complex ultrabornological PLS-spaces satisfying (PA) [1, Theorem 4.3].

5.13. **Remark.** It follows from Theorem 5.12 “ $d \Leftrightarrow e$ ” with $P(D) = \bar{\partial}$ and the fact that ultrabornological PLS-spaces are complete, in particular, quasi-complete that [20, Theorem 3.1, p. 989] (Dolbeaut–Grothendieck resolution of ${}^E\tilde{\mathcal{O}}^p$, cf. [21, 2.1.3 Theorem, p. 76]) is not correct for $p = n = 1$ and ultrabornological PLS-spaces E without (PA) , for instance, for the spaces E from Example 4.5 a).

Clearly, there are still some open problems.

- 5.14. **Problem.**
- (i) Is *strict admissibility* a necessary condition for the existence of a reasonable theory of E -valued Fourier hyperfunctions for general sequentially complete $\mathbb{C}$ -lcHs E ? In particular, does such a reasonable theory exist for the spaces E from Example 4.5 b)?
 - (ii) Are strict admissibility and admissibility equivalent?
 - (iii) Is strict admissibility of a sequentially complete E equivalent to belonging to the classes of spaces from Theorem 4.3?
 - (iv) Do the results for E -valued Fourier hyperfunctions in one variable ($d = 1$) carry over to several variables ($d \geq 2$)?

Figure 1: Problem 5.14 in the source paper, source PDF page 46.

2 Definitions used

A complex locally convex Hausdorff space E is *admissible* if, for every compact $K \subset \bar{\mathbb{R}}$, the Cauchy–Riemann operator

$$\bar{\partial} : \mathcal{E}^{\exp}(\bar{\mathbb{C}} \setminus K, E) \longrightarrow \mathcal{E}^{\exp}(\bar{\mathbb{C}} \setminus K, E)$$

is surjective. It is *strictly admissible* if it is admissible and

$$\bar{\partial} : C^\infty(U, E) \longrightarrow C^\infty(U, E)$$

is surjective for every open $U \subset \mathbb{C}$ [1, Definition 4.1].

Write $\mathfrak{c} = |\mathbb{R}| = |\mathbb{C}|$ for the cardinality of the continuum. Let I be any set with $|I| = 2^\mathfrak{c}$ and equip

$$E := \mathbb{C}^I$$

with its product locally convex topology.

3 Proof intuition

The analytic property in the question is completely coordinatewise for a product. A smooth map into $\mathbb{C}^I$ is precisely a family of scalar smooth maps, and every continuous seminorm on the product sees only finitely many coordinates. Scalar $\bar{\partial}$ -solutions can therefore be chosen independently in every coordinate without losing any weighted estimate. This makes $\mathbb{C}^I$ strictly admissible for an arbitrary index set I .

The rest of the argument chooses I deliberately large and separates the resulting product from the three classes by elementary invariants. It is not metrizable; it has no countable fundamental family of bounded sets, while a strong dual of a Fréchet space does; and its underlying cardinality is larger than the continuum, while every PLS-space has cardinality at most the continuum. Completeness is preserved by arbitrary products.

4 Formal result

Lemma 1 (Products preserve strict admissibility). *Let $(E_j)_{j \in J}$ be any family of strictly admissible complex locally convex Hausdorff spaces. Then $\prod_{j \in J} E_j$, with the product topology, is strictly admissible. If every E_j is complete, then the product is complete.*

Proof. Put $X = \prod_{j \in J} E_j$ and denote the coordinate projections by π_j . A map into a locally convex product is smooth exactly when all its coordinate maps are smooth, with derivatives computed coordinatewise. Moreover, every continuous seminorm p on X is dominated by

$$C \max_{j \in F} p_j \circ \pi_j$$

for some finite $F \subset J$, some continuous seminorms p_j on E_j , and some $C > 0$.

Fix a compact $K \subset \mathbb{R}$ and $f \in \mathcal{E}^{\exp}(\mathbb{C} \setminus K, X)$. For every j , the coordinate $f_j = \pi_j \circ f$ belongs to $\mathcal{E}^{\exp}(\mathbb{C} \setminus K, E_j)$. Admissibility of E_j gives u_j in that same space with $\bar{\partial}u_j = f_j$. Define $u = (u_j)_{j \in J}$. Coordinatewise smoothness makes u smooth. The finite coordinate domination of every continuous seminorm shows, directly in each defining weighted seminorm, that $u \in \mathcal{E}^{\exp}(\mathbb{C} \setminus K, X)$. Hence $\bar{\partial}u = f$.

The identical coordinatewise construction, without weights, solves $\bar{\partial}u = f$ for every $f \in C^\infty(U, X)$ and every open $U \subset \mathbb{C}$. Thus X is strictly admissible. Finally, an arbitrary product of complete Hausdorff uniform spaces is complete, so the last claim follows. $\square$

Lemma 2 (Two size obstructions). *Let I be uncountable.*

1. *The product $\mathbb{C}^I$ is not metrizable and has no countable fundamental family of bounded sets.*
2. *Every PLS-space has underlying cardinality at most $\mathfrak{c}$.*

Proof. For the first assertion, suppose (V_n) were a countable neighborhood base at zero. Each product neighborhood restricts only finitely many coordinates. The union of those finite coordinate sets is countable, so one may choose $i \in I$ outside it. Every V_n then contains the whole i -th coordinate line, whereas the neighborhood $\{x : |x_i| < 1\}$ does not. This is a contradiction.

A subset of $\mathbb{C}^I$ is bounded exactly when it is bounded in every coordinate. Given any sequence (B_n) of bounded sets, choose distinct $i_n \in I$, put

$$M_n := \sup_{x \in B_n} |x_{i_n}| < \infty,$$

and define $x \in \mathbb{C}^I$ by $x_{i_n} = M_n + 1$ and $x_i = 0$ on all other coordinates. The singleton $\{x\}$ is bounded, but $x \notin B_n$ for every n . Hence no countable sequence of bounded sets is cofinal.

For the second assertion, recall that a DFS-space is the strong dual of a Fréchet–Schwartz space. Every Fréchet–Schwartz space is separable: in a Banach-step representation the relevant linking maps are compact with dense range, so each Banach step is separable, and the Fréchet space embeds into a countable product of separable metrizable spaces. A continuous functional on a separable Hausdorff topological vector space is determined by its values on a countable dense set. Therefore every DFS-space has cardinality at most $\mathfrak{c}$. A PLS-space is a projective limit of a sequence of DFS-spaces, hence is a subset of their countable product and has cardinality at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$. $\square$

Theorem 1 (Negative answer to Problem 5.14(iii)). *Strict admissibility of a sequentially complete complex locally convex Hausdorff space is not equivalent to membership in the three classes of Theorem 4.3 of [1].*

More explicitly, if $|I| = 2^{\mathfrak{c}}$, then $E = \mathbb{C}^I$ is complete and strictly admissible, but it is neither

- 1. a Fréchet space,*
- 2. a strong dual F'_b of a Fréchet space F satisfying (DN), nor*
- 3. an ultrabornological PLS-space satisfying (PA).*

Proof. The scalar field $\mathbb{C}$ is a complete strictly admissible space; this is also the one-dimensional instance of Theorem 4.3(a) in [1]. Lemma 1 therefore shows that $E = \mathbb{C}^I$ is complete, hence sequentially complete, and strictly admissible.

Because I is uncountable, Lemma 2(1) shows that E is not metrizable, so it is not Fréchet.

Suppose next that E were topologically isomorphic to F'_b for some Fréchet space F (the condition (DN) is not needed here). Let (U_n) be a countable zero-neighborhood base in F . Every bounded subset B of the strong dual is pointwise bounded and hence equicontinuous by Banach–Steinhaus, because F is barrelled. Thus $B \subset U^\circ$ for some zero-neighborhood U ; choosing n with $U_n \subset U$ gives $B \subset U^\circ \subset U_n^\circ$. Consequently (U_n°) is a countable fundamental family of bounded sets in F'_b . This contradicts Lemma 2(1).

Finally,

$$|E| = |\mathbb{C}|^{|I|} = \mathfrak{c}^{2^{\mathfrak{c}}} = (2^{\aleph_0})^{2^{\mathfrak{c}}} = 2^{2^{\mathfrak{c}}} > \mathfrak{c}.$$

Lemma 2(2) therefore rules out even that E is a PLS-space, a fortiori that it belongs to class (c). This excludes all three classes and proves the result. $\square$

5 Verification notes

- **Hypotheses.** The example is a complex locally convex Hausdorff space and is complete, not merely sequentially complete.
- **Analytic step.** Both weighted and unweighted $\bar{\partial}$ equations are checked coordinatewise. The only topology fact used is that continuous seminorms on a product depend on finitely many coordinates.
- **Class (b).** The obstruction is stronger than required: the example is not the strong dual of any Fréchet space, regardless of (DN).

- **Class (c).** The cardinal argument rules out the PLS property itself, before (PA) or ultrabornologicity is considered.
- **Set theory.** The choice $|I| = 2^{\mathfrak{c}}$ and the strict inequality $2^{2^{\mathfrak{c}}} > \mathfrak{c}$ use only Cantor’s theorem and are valid in ZFC.

6 Scope and novelty check

This settles the literal form of Problem 5.14(iii) negatively. It does not address Problems 5.14(i), (ii), or (iv). The counterexample is intentionally very large and nonmetrizable. If the intended question was restricted to an unstated smaller category of “natural” locally convex spaces, that narrower variant remains open and should be distinguished from the printed statement.

A bounded novelty check was performed on 27 August 2026. The four local run indexes were searched for arXiv:1912.03659 and the phrases “strict admissibility”, “admissible”, “Fourier hyperfunctions”, and “product”. Web/arXiv searches used the exact wording of Problem 5.14(iii), the paper title and author, and the phrases “strictly admissible product” and “classes of spaces from Theorem 4.3”. The current arXiv v2 source, dated 5 November 2025, still states the problem. No separate paper or packet asserting this product counterexample or answering Problem 5.14(iii) was found. This is a bounded, not exhaustive, novelty search; novelty confidence is therefore moderate.

7 Human-review recommendation

Promote as a full negative answer to Problem 5.14(iii) if a reviewer confirms (1) that the problem is intended literally over all sequentially complete complex locally convex Hausdorff spaces, and (2) the standard cardinal bound $|X| \leq \mathfrak{c}$ for PLS-spaces. The coordinatewise strict-admissibility argument and the strong-dual bounded-set obstruction are otherwise elementary.

References

- [1] K. Kruse, *Vector-valued Fourier hyperfunctions and boundary values*, arXiv:1912.03659v2 (5 November 2025), <https://arxiv.org/abs/1912.03659>; published in *Kyoto Journal of Mathematics*, doi:10.1215/21562261-2024-0030.